THE UNITED REPUBLIC OF TANZANIA
THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT
ANNUAL JOINT EXAMINATION

FORM THREE
BASIC MATHEMATICS
MARKING SCHEME NOVEMBER 2024

SECTION A

1. (a) i. $1\ 109^2 = 1\ 229\ 881$..... $00\frac{1}{2}$ mark

$\sqrt{841} = \pm 29$..... $00\frac{1}{2}$ mark

The least or lowest sum $= 1\ 229\ 881 - 29 = 1\ 229\ 852$..... $00\frac{1}{2}$ mark

$1\ 229\ 852 =$ One million, two hundred twenty nine thousand, eight hundred fifty two. $00\frac{1}{2}$ mark

ii. $0.454545 \ldots = 0.\overline{45}$

Let $0.\overline{45}\ be\ y$

$y = 0.\overline{45}$.....equation 1

$y \times 100 = 0.\overline{4545} \times 100 = 45.45$

$100y = 45.45$ equation2

Subtract equation 1 from equation 2

$$-\left|\begin{matrix}100y = 45.45\\ y = 0.45\end{matrix}\right.$$

$99y = 45$

$\frac{99y}{99} = \frac{45}{99}\ \ y = \frac{45\div 9}{99\div 9} = \frac{5}{11}$.....01 mark.

Fraction for girls $= 1 - \frac{5}{11} = \frac{6}{11}$.....01 mark

(b) i. $0.0607879 \approx 0.061$ ii. $0.0607879 \approx 0.0608$ .....02 marks

2. (a) $100^{\sqrt{x}} = \sqrt{1\ 000^x}$

$10^{2\sqrt{x}} = \sqrt{10^{3x}}$

$10^{2\sqrt{x}} = (10^{3x})^{\frac{1}{2}}$

$10^{2\sqrt{x}} = 10^{\frac{3x}{2}}$

$2\sqrt{x} = \frac{3x}{2}$.....01 mark

$\left(2\sqrt{x}\right)^2 = \left(\frac{3x}{2}\right)^2$

$4x = \frac{9x^2}{4}$.....01 mark

$\frac{9x^2}{4} - 4x = 0$

$x\left(\frac{9x}{4} - 4\right) = 0$

Either $x = 0\ or\ \frac{9x}{4} - 4 = 0$

Therefore, $x = 0\ or\ x = \frac{16}{9}$ .....01 mark

(b) i. $M = \log_2\big((x+3)(x-3)\big)$

$= \log_2(x^2 - 9)$

$2^M = x^2 - 9$ ..... $00\frac{1}{2}$ mark.

$x^2 = 2^M + 9$

$x = \pm\sqrt{2^M + 9}$ .....01 mark

ii. From $M = 4$, $x = \pm\sqrt{2^4 + 9}$

$= \pm\sqrt{16 + 9}$

$= \pm\sqrt{25}$.....$00\frac{1}{2}$ mark

$= \pm 5$

$x = 5, x \geq 0$ .....01 mark

3. (a) $n(U) = 20,$

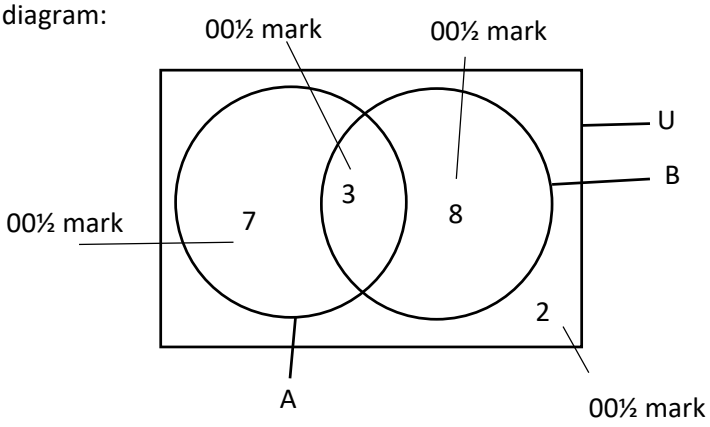
$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$n(A \cap B) = n(A) + n(B) - n(A \cup B)$

$= 10 + 11 - 18$

$= 3$

From Venn diagram:



$n(A \cap B)' = 7 + 2 + 8 = 17$.....01 mark

(b) Number of subsets in a set is given by 2^n ..... $00\frac{1}{2}$ mark

$2^n = 128$.....$00\frac{1}{2}$ mark

$2^n = 2^7$ .....01 mark

$n = 7$

Therefore, there are 7 elements.....01 mark

4. (a) i. Solve for y from $2x - 3y + 7 = 0$

$3y = 2x + 7$.....01 mark

$\frac{3y}{3} = \frac{2x+7}{3}$.....01 mark.

$y = \frac{2x}{3} + \frac{7}{3}$.....01 mark

ii. Gradient or slope $= m = \frac{2}{3}$.....01 mark

iii. $y-$ Intercept $= c = \frac{7}{3}$.....01 mark

(b) $m = \frac{y_2 - y_1}{x_2 - x_1}$.....$00\frac{1}{2}$ mark

$2 = \frac{4-2}{4-x}$ .....$00\frac{1}{2}$ mark

$2(4 - x) = 2$

$8 - 2x = 2$.....$00\frac{1}{2}$ mark

$8 - 2 = 2x$

$6 = 2x$.....$00\frac{1}{2}$ mark

$\frac{2x}{2} = \frac{6}{2}$

$x = 3$.....01 mark

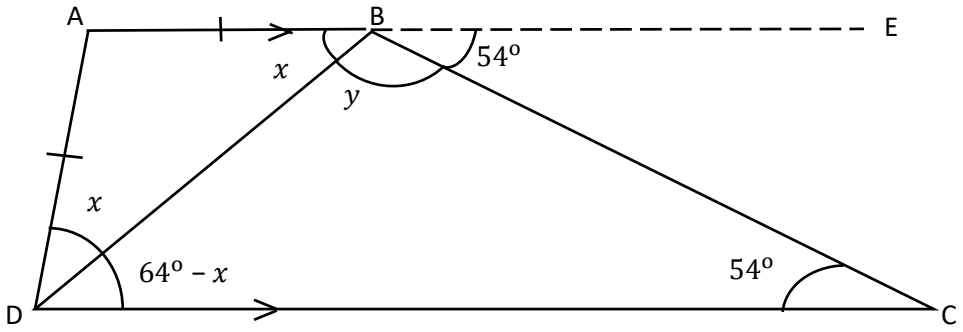
5. (a) $\overline{BF} = \overline{CE}$ (given side)..... $00\frac{1}{2}$ mark

$\angle BEC = \angle BFC = 90^{\circ}$ (given angle)..... $00\frac{1}{2}$ mark

$\overline{BC}$ is common side(hypotenuse).....01 mark

Therefore $\overline{CF} = \overline{BE}$ by RHS postulate.....01 mark

(b) Extend line $\overline{AB}$ *to* E



$\angle DCB = \angle EBC\ (\overline{AB} \parallel \overline{DC}\ alternate\ interior\ angles)$.....$00\frac{1}{2}$ mark

$\angle ADB = \angle ABD(Base\ angles\ of\ isoscless\ triangle)$ ..... $00\frac{1}{2}$ mark

Let $\angle DBC$ *be* y *and* $\angle ABD$ *be* x

$x + y + 54^{\circ} = 180^{\circ}$ (Straight-line angle)

$x + y = 180^{\circ} - 54^{\circ}$

$x + y = 126^{\circ}$ .....equation 1 ..... $00\frac{1}{2}$ mark

$\angle BDC = 64^{\circ} - x$

m$64^{\circ} - x + 54^{\circ} + y = 180^{\circ}$

$y - x = 180^{\circ} - (64^{\circ} + 54^{\circ})$

$y - x = 62^{\circ}$.....equation 2..... $00\frac{1}{2}$ mark

Solve equation 1 and 2 by elimination method

$$+\left|\begin{matrix} y + x = 126^{\circ} \\ y - x = 62^{\circ} \end{matrix}\right.$$

$2y = 188^{\circ}$ ..... $00\frac{1}{2}$ mark

$\frac{2y}{2} = \frac{188^{\circ}}{2}$

$y = \angle DBC = 94^{\circ}$.....$00\frac{1}{2}$ mark

6. (a) 59 dag= 590 g.....01 mark

$$-\left|\begin{matrix} 112\ kg & 70\ g \\ 35\ kg & 590\ g \end{matrix}\right.$$ .....01 mark

76 kg 480 g

Therefore the mother weighs 76 kg 480 g.....01 mark

(b) Let Seypalm's speed be $x\ km|\ hr\ and\ Jonathan's\ speed\ be\ y\ km|hr$

From time = $\frac{distance}{speed}$, then

$\frac{30}{x} - \frac{30}{y} = 2$.....equation 1 ..... $00\frac{1}{2}$ mark

$\frac{30}{y} - \frac{30}{2x} = 1$.....equation 2..... $00\frac{1}{2}$ mark

Solve equation 1 and 2 by elimination method

$\frac{30}{x} - \frac{30}{2x} = 3$..... $00\frac{1}{2}$ mark.

$x\left(\frac{30}{x} - \frac{30}{2x} = 3\right)$

$30 - 15 = 3x$

$3x = 15$ ..... $00\frac{1}{2}$ mark

$\frac{3x}{3} = \frac{15}{3}$

$x = 5$

Therefore, Seypalm's speed is 5 km|hr .....01 mark

7. (a) $\frac{x}{y} = \frac{0.04}{1.5} = \frac{4}{150} = \frac{2}{75}$ .....01 mark

$= \frac{y-x}{x+y} = \frac{\frac{y}{y}-\frac{x}{y}}{\frac{x}{y}+\frac{y}{y}}$.....$00\frac{1}{2}$ mark

$= \frac{1-\frac{x}{y}}{\frac{x}{y}+1} = \frac{1-\frac{2}{75}}{1+\frac{2}{75}}$.....$00\frac{1}{2}$ mark

$= \frac{\frac{75-2}{75}}{\frac{77}{75}} = \frac{\frac{73}{75}}{\frac{77}{75}}$

$= \frac{73}{75} \times \frac{75}{77} = \frac{73}{77}$.....01 mark.

(b) THE TRIAL BALANCE OF SEYPALM CO BUSINESS

Name of account	Debit	Credit
Capital		30 000
Furniture	25 000	
Motor vehicle	45 000	
Sales		68 000
Purchases	54 000	
Creditor		76 000
Debtor	15 000	
Insurance	3 000	
Cash	18 000	
Discount received		7 000
Discount allowed	4 000	

Drawing	12 000	
Electricity	5 000	
	181 000	181 000.....03 marks

8. (a) Let 5 arithmetic mean numbers be $a, b\ c, d. e$

$A_1 = 17$

$A_2 = A_1 + d = a,\ A_3 = A_1 + 2d = b, A_4 = A_1 + 3d = c$

$A_5 = A_1 + 4d = d\ , A_6 = A_1 + 5d = e, A_7 = A_1 + 6d = 53$

From $A_7 = A_1 + 6d = 53$ but $A_1 = 17\ \ 17 + 6d = 53$

$6d = 53 - 17 = 36$

$\frac{6d}{6} = \frac{36}{6}\ , d = 6$ ..... $00\frac{1}{2}$ mark

$A_2 = a = 17 + 6 = 23$..... $00\frac{1}{2}$ mark

$A_3 = b = 17 + 2 \times 6 = 17 + 12 = 29$..... $00\frac{1}{2}$ mark

$A_4 = c = 17 + 3 \times 6 = 35$..... $00\frac{1}{2}$ mark

$A_5 = d = 17 + 4 \times 6 = 41$ ..... $00\frac{1}{2}$ mark

$A_6 = e = 17 + 5 \times 6 = 47$..... $00\frac{1}{2}$ mark

Therefore, the 5 arithmetic mean between 17 and 53 are 23, 29, 35, 41, and 47

(b) $G_1 = p - 2, G_2 = p - 1, G_3 = 3p - 5$

$r = \frac{G_3}{G_2} = \frac{G_2}{G_1}$

$r = \frac{3p-5}{p-1} = \frac{p-1}{p-2}$

$(3p - 5)(p - 2) = (p - 1)(p - 1)$

$3p^2 - 6p - 5p + 10 = p^2 - 2p + 1$

$3p^2 - p^2 - 11p + 2p + 10 - 1 = 0$

$2p^2 - 9p + 9 = 0$.....01 mark

Solve the equation by general formula

$p = \frac{-b \pm \sqrt{b^2-4ac}}{2a}\ , where, a = 2, b = -9, c = 9$

$= \frac{9 \pm \sqrt{9^2-4\times2\times9}}{2\times2}$

$= \frac{9 \pm \sqrt{81-72}}{4}$

$= \frac{9 \pm \sqrt{9}}{4} = \frac{9 \pm 3}{4}$

$= \frac{9+3}{4}\ or\ \frac{9-3}{4}$

$= \frac{12}{4}\ or\ \frac{6}{4}$

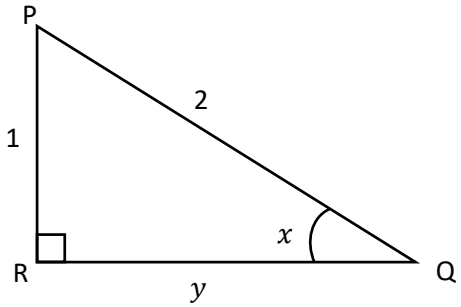
$= 3\ or\ \frac{3}{2}$.....02 marks

9. (a) i. Sin $x = \frac{2y}{4y} = \frac{1}{2}$.....01 mark

$x = sin^{-1}\left(\frac{1}{2}\right)$

$= 30^{\circ}$.....01 mark

ii. From triangle PQR



01 mark

$y = \sqrt{2^2 - 1^2} = \sqrt{4-1} = \sqrt{3}$ ....01 mark

$tanx = \frac{1}{\sqrt{3}} = \frac{1}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$

$= \frac{\sqrt{3}}{3}$.....01 mark

(b) $diagonal = \sqrt{side(s)^2 + side\ (s)^2}$

$= \sqrt{(9\ cm)^2 + (9\ cm)^2}$.....$00\frac{1}{2}$ mark.

$= \sqrt{81\ cm^2 + 81\ cm^2}$.....$00\frac{1}{2}$ mark

$= \sqrt{162\ cm^2}$.....01 mark

$= \left(9\sqrt{2}\right) cm$.....01 mark

10. (a) i. $\frac{(67.542)^2 - (32.458)^2}{75.458 - 40.374} = \frac{(67.542 - 32.458)(67.542 + 32.458)}{34.718}$.....$00\frac{1}{2}$ mark

$= \frac{35.084 \times 100}{35.084}$.....$00\frac{1}{2}$ mark

$= 100$..... $00\frac{1}{2}$ mark

ii. $\frac{2x}{1+\frac{1}{1+\frac{x}{1-x}}} = 1$

$\frac{2x}{1+\frac{1}{\frac{x+1-x}{1-x}}} = 1$

$\frac{2x}{1+\frac{1}{\frac{1}{1-x}}} = 1$

$\frac{2x}{1+1-x} = 1$

$\frac{2x}{2-x} = 1$..... $00\frac{1}{2}$ mark

$2x = 2 - x$

$2x + x = 2$

$3x = 2$..... $00\frac{1}{2}$ mark

$\frac{3x}{3} = \frac{2}{3}$

$x = \frac{2}{3}$.....$00\frac{1}{2}$ mark

(b) $(2p-1)(p+1) = 2(2p+2)$

$2p^2 + 2p - p - 1 = 4p + 4$

$2p^2 + 2p - p - 4p - 1 - 4 = 0$

$2p^2 - 3p - 5 = 0$.....01 mark

Solve the equation by method of factorization

Product is -10 and factors are $(-5, 2)$ .....01 mark

$2p^2 - 5p + 2p - 5 = 0$

$p(2p-5)1(2p-5) = 0$

Either $(p+1) = 0$ *or* $(2p-5) = 0$

$p = -1$ *or* $p = \frac{5}{2}$.....01 mark

SECTION B

11. (a) So $= x - 2, OQ = OP = x$ $(radii)$.....$00\frac{1}{2}$ mark

$OP^2 = OS^2 + SP^2$

$x^2 = (x-2)^2 + 6^2$.....$00\frac{1}{2}$ mark

$x^2 = x^2 - 4x + 4 + 36$

$4x = 40$.....01 mark

$\frac{4x}{4} = \frac{40}{4}$

$x = 10$ .....01 mark

(b)

Class interval	cf	f	cf	Class mark(x)	fx_i	Upper real class limit
$125 - 130$	5	5	5	127.5	635.5	130.5
$131 - 136$	13	8	13	132.5	1 060	136.5
$137 - 142$	25	12	25	137 .5	1 650	142.5
$143 - 148$	29	4	29	142.5	570	148.5
$149 - 154$	30	1	30	147. 5	147.5	154.5
Total		30				
		... $00\frac{1}{2}$	$00\frac{1}{2}$ mark	.. $00\frac{1}{2}$ mark	Σfx_i $= 4\,063$ ... 01 mark	.. $00\frac{1}{2}$ mark

i. $\overline{x} = \frac{\Sigma fx_i}{\Sigma f_i} = \frac{4\,063}{30}$..... $00\frac{1}{2}$ mark

$= 135.43$..... .. $00\frac{1}{2}$ mark

ii. Modal class $137 - 142$

$Mode = l + \left(+\frac{t_1}{t_1 + t_2}\right) i$

$= 136.5 + \left(\frac{12-8}{(12-8)+(12-4)}\right) 5$

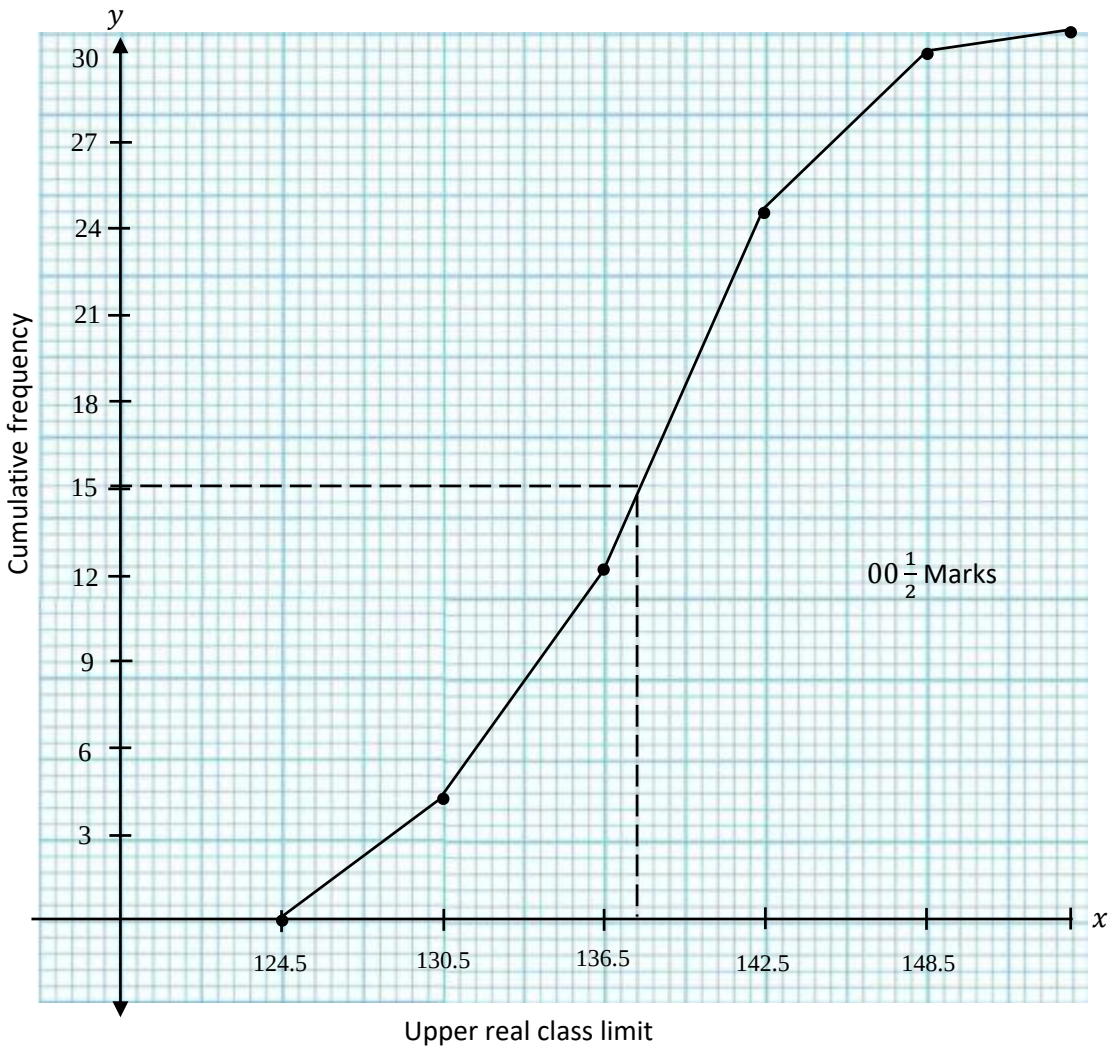
$= 136.5 + \left(\frac{4}{4+8}\right) 5$..... $00\frac{1}{2}$ mark

$= 136.5 + \frac{4}{12} \times 5$

$= 136.5 + 1.66666..$

$= 138.17$.....$00\frac{1}{2}$ mark

iii. Ogive graph or curve



Median position $= \left(\frac{n}{2}\right)^{th} = \left(\frac{30}{2}\right)^{th} = 15^{th}$

$= \frac{6}{10} \times 2 = 1.2$

$= 136.5 + 1.2$

$= 137.7$

Therefore, estimated median is 137.7 cm..... $00\frac{1}{2}$ mark

12. (a) Angular distance $= 39^{\circ} - 33^{\circ} = 6^{\circ}$.....01 mark.

Since latitudes are constant, therefore the points are on the small circle

Distance in nautical miles $= 6^{\circ} \times 60' cos 6^{\circ}$ ..... 01 mark

$= 358.028\, nm$ .....01 mark

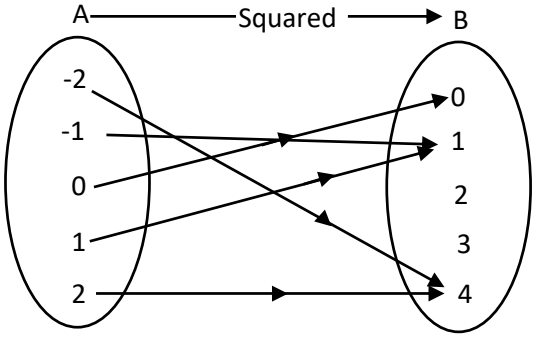
$t = \frac{distance}{speed}$

$= \frac{358.028'}{20 nautical\ miles\ per\ hour}$ .....01 mark

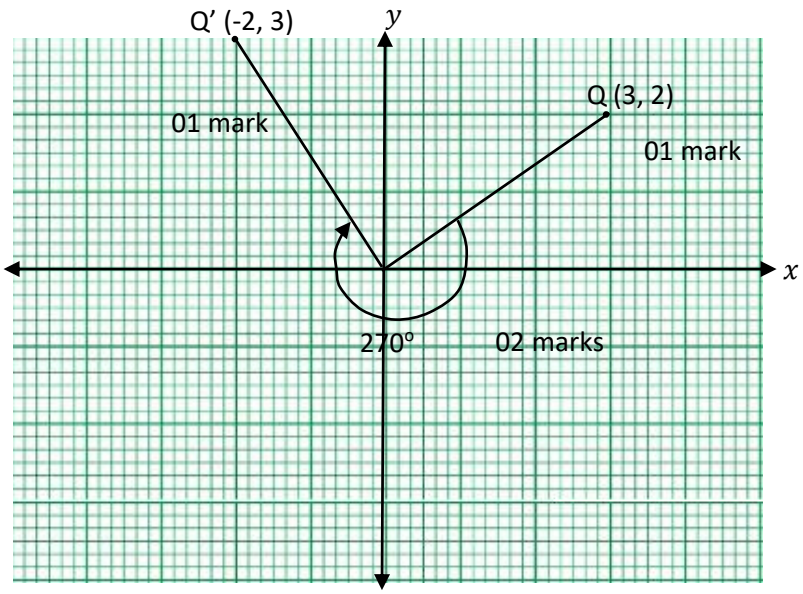
$= 17.9\ hours$

Therefore, it will reach at point B after 17.9 hours .....01 mark

(b) Diagram ……05 marks



13. (a) The rotation of point Q (3, 2) through 270° clockwise about the origin is shown in the following figure



From the figure above $R_{270^{\circ}}(3\ ,2) = (-2\ ,3)$.....01 mark

Generally, the rotation of a point through 270° clockwise about the origin is given by $R_{270^{\circ}}(x, y) = (-x\ , y)$

(b) Principal (p) $= 5\ 000\ 000$ shillings

Amount (A)$= 10\ 000\ 000$ shillings

Number of times interest is compounded per year (n) $= 2$

Rate (r) $= 5.5\%$

Required : time (t)

$$A_t = p\left(1 + \frac{r}{100n}\right)^{nt}$$

$$10\ 000\ 000 = 5\ 000\ 000\left(1 + \frac{5.5}{200}\right)^{2t}$$

$\frac{10\ 000\ 000}{5\ 000\ 000} = \frac{5\ 000\ 000}{5\ 000\ 000}(1.0275)^{2t}$.....$00\frac{1}{2}$ mark

$2 = (1.0275)^{2t}$ introduce log both sides.....01 mark

$\log 2 = \log(1.0275)^{2t}$.....$00\frac{1}{2}$ mark

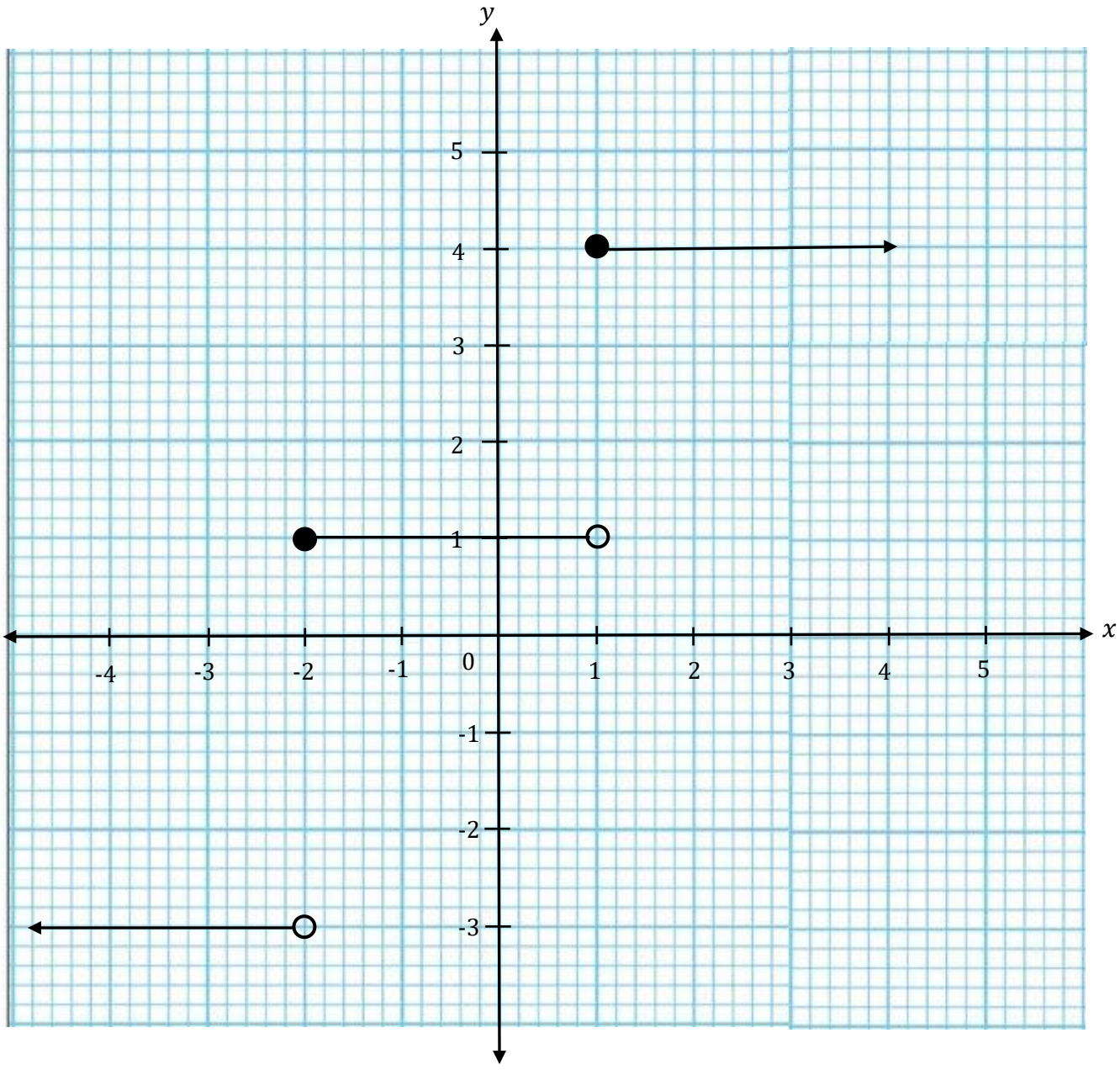
$2t = \frac{\log 2}{\log 1.0275}$.....01 mark

$t = \frac{1}{2}(25.55035862 \ldots) = 12.775.. \approx 13$ years.....01 mark

Therefore, it will double after 13 years.....01 mark

14. (a) $y = -3\ for = \cdots -4, -3, -2$
$y = 1\ for\ x = -2, -1, 0\ , 1$
$y = 4\ for\ x = 1\ , 2\ , 3\ , \ldots$

i.



..... 02 marks

ii. Domain $= \{All\ real\ numbers\}$.....01 marks

iii. $Range = \{-3, 1, 4\}$.....01 marks

iv. It is not one to one function.....01 mark

(b) Table of values

x	...	-3	-2	-1	0	1	2	3	...
$y = -x^2 + 2$	...	-7	-2	1	2	1	-2	-7	...

.....02 marks

Domain= $\{all\ real\ numbers\}$.....00$\frac{1}{2}$ mark

Range= $\{y: y \leq 2\}$.....00$\frac{1}{2}$ mark

